

Sai Rakshith Muda

State College, PA | (814) 780-1228 | rakshithmuda@gmail.com

github.com/rakshithmuda22 | linkedin.com/in/sai-rakshith-muda-09341a1b4 | rakshithmuda22.github.io

EDUCATION

Pennsylvania State University

State College, PA

Bachelor of Science in Computer Science, GPA 3.45

May 2026

- Coursework: Operating Systems, Artificial Intelligence, Database Management Systems, Communication Networks, Theory of Computation

TECHNICAL SKILLS

Languages: Python, TypeScript, Dart, Swift, C, SQL

AI and LLM: Anthropic Claude API, Groq API, LangGraph, LangChain, Retrieval-Augmented Generation (RAG), ChromaDB, sentence-transformers

Backend: FastAPI, Flask, Pydantic, Supabase, PostgreSQL, Redis, MySQL, REST APIs

Frontend: React, Flutter, SwiftUI, Tailwind CSS

Infrastructure and DevOps: Docker, GitHub Actions, CI/CD, Cloudflare Workers, Vercel, Deno

Testing and Tools: Git, pytest, asyncio, tree-sitter, Redis Queue (RQ), webhooks

PROJECTS

AgentSpec | Python, Pydantic, pytest

github.com/rakshithmuda22/agentspec

- A pip-installable library (`pip install llm-agentspec`) that checks whether an LLM agent called the right tools in the right order, using rules such as `must_call`, `must_call_before`, and `must_not_call`. It gives a deterministic pass or fail on agent behavior, which output-grading tools like DeepEval and RAGAS do not.
- Runs offline in Python with no API keys. A framework-agnostic `AgentSession` reads traces from Anthropic `tool_use` blocks, and one `assert_all_pass()` call drops the checks into an existing pytest suite for CI. Covered by 87 tests.

AgentProbe | Python, LangGraph, FastAPI

github.com/rakshithmuda22/agentprobe

- A GitHub Action that reviews pull requests against a team's architectural boundaries and naming rules using three LangGraph-orchestrated agents (Architecture, Pattern, Regression), and blocks the merge when a boundary is crossed.
- Regression detection runs in two layers, Logic Summary Diffing and Property-Based Testing, with weighted scoring. Backed by 90 tests including HMAC-SHA256 webhook verification; teams set boundaries and forbidden patterns in YAML.

CodeSage | Python, FastAPI, Groq, Redis

github.com/rakshithmuda22/codesage

- Four agents (BugDetector, SecurityScanner, StyleAdvisor, TestCoverage) review a pull request in parallel with `asyncio.gather()` and post inline comments at exact diff positions on GitHub. StyleAdvisor checks code against each repo's own conventions, learned from its last three merged PRs, and parses source with tree-sitter across Python, JavaScript, and TypeScript. 38 tests.

Stacktrace-to-PR Time Machine | Python, FastAPI

github.com/rakshithmuda22/stacktrace-time-machine

- A stateless webhook service that takes a Sentry error and traces it back to the pull request that introduced the bug, using GitHub's GraphQL blame API with no local git clones. Queries pin to the deployed release SHA so line numbers do not drift, and it blames the whole enclosing function by parsing Python with `ast`. 64 tests.

DocuMind | Python, FastAPI, ChromaDB, RAG

github.com/rakshithmuda22/Documind

- A retrieval-augmented generation chatbot that answers questions over uploaded PDFs and cites its sources, using PyMuPDF parsing, all-MiniLM-L6-v2 embeddings, ChromaDB search, and Groq LLaMA 3.1. Containerized with Docker for Hugging Face Spaces; LLM calls are mocked so the 44 tests run in CI without keys.

EXPERIENCE

Exam Proctor, Mathematics Department

Mar 2024 - Dec 2025

Pennsylvania State University

State College, PA

- Ran exam logistics and kept testing environments secure; verified student identities and handled confidential examination materials.

Front Desk Operations, Commons

Mar 2024 - Apr 2026

Pennsylvania State University

State College, PA

- Recognized as Employee of the Month. Managed student information systems and followed campus check-in and security procedures.

CERTIFICATIONS

Introduction to DevOps, IBM (Coursera), Jan 2026 | Claude 101, Claude Code in Action, and Introduction to Agent Skills, Anthropic Academy, 2026